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Experiment 6:

Experiment Aim: Learn how to create, query, and manage views in SQL to simplify database queries and provide a layer of abstraction for end-users. (Company Tags: Amazon, Zoho, ServiceNow)

Tools used: PostgreSQL

Objectives

- **Data Abstraction:** To understand how to hide complex table joins and calculations behind a simple virtual table interface.
- **Enhanced Security:** To learn how to restrict user access to sensitive columns by providing views instead of direct table access.
- **Query Simplification:** To master the creation of views that pre-join multiple tables, making reporting easier for non-technical users.
- **View Management:** To understand the syntax for creating, altering, and dropping views, as well as the naming conventions required for efficient data access.

Experiment Steps:

Table Creation and Data Storing-

```
CREATE TABLE Departments (
   DeptID INT PRIMARY KEY,
   DeptName VARCHAR(50)
);

CREATE TABLE Employees (
   EmpID INT PRIMARY KEY,
   EmpName VARCHAR(50),
   DeptID INT,
   Salary NUMERIC(10,2),
   Status VARCHAR(10),
   FOREIGN KEY (DeptID) REFERENCES Departments(DeptID)
);

INSERT INTO Departments VALUES
(1,'IT'),
(2,'HR'),
(3,'Finance');

INSERT INTO Employees VALUES
(101,'Amit',1,50000,'Active'),
(102,'Neha',2,45000,'Inactive'),
(103,'Rahul',1,60000,'Active'),
(104,'Pooja',3,70000,'Active');

select * from Departments;
```

	deptid [PK] integer	deptname character varying (50)
1	1	IT
2	2	HR
3	3	Finance

select * from Employees;

	empid [PK] integer	empname character varying (50)	deptid integer	salary numeric (10,2)	status character varying (10)
1	101	Amit	1	50000.00	Active
2	102	Neha	2	45000.00	Inactive
3	103	Rahul	1	60000.00	Active
4	104	Pooja	3	70000.00	Active

Step 1: Creating a Simple View for Data Filtering

Implementing a view to provide a quick list of active employees without exposing the entire table structure.

```
CREATE VIEW ActiveEmployees AS
SELECT EmpID, EmpName, DeptID, Salary
FROM Employees
WHERE Status = 'Active';
```

select * from ActiveEmployees;

	empid integer	empname character varying (50)	deptid integer	salary numeric (10,2)
1	101	Amit	1	50000.00
2	103	Rahul	1	60000.00
3	104	Pooja	3	70000.00

Step 2: Creating a View for Joining Multiple Tables

Simplifying the retrieval of data distributed across Employees and Departments tables.

```
CREATE VIEW EmpDeptView AS
SELECT
    e.EmpID,
    e.EmpName,
    d.DeptName,
    e.Salary
FROM Employees e
JOIN Departments d ON e.DeptID = d.DeptID;
```

```
SELECT * FROM EmpDeptView;
```

	empid integer	empname character varying (50)	deptname character varying (50)	salary numeric (10,2)
1	101	Amit	IT	50000.00
2	102	Neha	HR	45000.00
3	103	Rahul	IT	60000.00
4	104	Pooja	Finance	70000.00

Step 3: Advanced Summarization

View Creating a view to provide department-level statistics automatically

```
CREATE VIEW DeptSalaryStats AS
SELECT
    d.DeptName,
    COUNT(e.EmpID) AS TotalEmployees,
    SUM(e.Salary) AS TotalSalary,
    AVG(e.Salary) AS AvgSalary
FROM Employees e
JOIN Departments d ON e.DeptID = d.DeptID
GROUP BY d.DeptName;
```

```
SELECT * FROM DeptSalaryStats;
```

	deptname character varying (50)	totalemployees bigint	totalsalary numeric	avgsalary numeric
1	Finance	1	70000.00	70000.000000000000
2	IT	2	110000.00	55000.000000000000
3	HR	1	45000.00	45000.000000000000

Outcomes:

- **Abstraction Proficiency:** Students will be able to create and query views to simplify efficient data access and abstraction.
- **Security Implementation:** Students will understand how to use views for data masking and providing restricted access to sensitive information.
- **Syntactic Accuracy:** Students will demonstrate the correct syntax for creating and querying views, ensuring logical clarity in naming conventions.
- **Real-world Application:** Students will be able to design views for practical domains like Library Management Systems or Payroll Systems to demonstrate functionality.